MIL-C-43899A(GL)
27 March 1985
SUPERSEDING
MIL-C-43899(GL)
28 June 1974

MILITARY SPECIFICATION

COUPLING, EXTRACTION FORCE TRANSFER, PLATFORM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This document covers one type of airdrop platform extraction force transfer coupling assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	-	Twine, Fibrous Cotton, Wrapping
PPP-B-26	-	Bag, Plastic, (General Purpose)
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-H-6875	-	Heat Treatment of Steel, Process for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1670

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

11-1-1674 - Coupling, Extraction Force Transfer, Platform; Assembly
11-1-1675 - Actuator, L.H.
11-1-1685 - Link Assembly
11-1-1686 - Bracket, Support
11-1-1687 - Tube Assemblies
11-1-1689 - Bridles
11-1-1692 - Actuator, R.H.

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish a sample for first article inspection (see 4.3, 6.2 and 6.3).

3.2 Materials (see 6.3) and components. Components of the coupling assembly shall conform to the materials, construction and supplementary process requirements shown or specified on the applicable assembly drawings listed below, detail drawings subsidiary thereto, and as specified herein.

Component	Drawings
Actuator, L.H.	11-1-1675
Actuator, R.H.	11-1-1692
Tubes	11-1-1687
Bridles	11-1-1689
Link	11-1-1685
Bracket	11-1-1686

Except that field assembly as shown on Drawing 11-1-1674 is not required, the complete coupling assembly shall consist of the components specified thereon in the quantity indicated.

3.2.1 Measuring and cutting of webbing. Webbing shall be measured and cut while substantially free of tension. Rolls of tightly wound webbing should be unwound and the webbing allowed to relax for a minimum of 12 hours prior to use.

3.2.2 Machine stitching. Thread breaks, bobbin runouts, or two or more consecutive skipped stitches shall be overstitched a minimum of 1 inch beyond ends of each defect. Runoff areas shall be restitched to conform to the required stitch pattern. Thread tension shall be maintained so that there will be no loose stitching and the lock will be imbedded in the material sewed. All thread ends shall be trimmed.

3.3 Performance, actuator assemblies.

3.3.1 Proofload. Actuator assemblies shall show no evidence of release, failure, or visible permanent deformation when subjected to a 9000 pound proofload as specified in 4.5.

3.3.2 Function. When tested as specified in 4.5, actuators shall function so as to effect positive release of the captive bridle. There shall be no evidence of binding or failure of metal components.

* 3.4 Marking. All parts that can be functionally disassembled shall be marked in accordance with MIL-STD-130.

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3.5 Workmanship. Items shall be processsed in such a manner as to be uniform in quality and shall be free from burrs, cracks, cutting edges, and other defects that will adversely affect life, serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test items to determine the validity of the certification.

* 4.2 Classification of inspection.

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3.1 and 4.4.3.2 and tested for the characteristics specified in 4.5. The presence of any defect or failure of any test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process examination. Inspection shall be made during manufacturing to determine that operations or assemblies are in accordance with the applicable drawings. Whenever nonconformance is noted, correction shall be made to the items affected and the lot in-process. Items which cannot be corrected shall be removed from production.

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4.4.2.1 Heat treatment examination. Heat treated components shall be inspected to determine compliance with requirements specified on the applicable drawings. The provisions of MIL-H-6875, applicable to routine quality control, shall apply, except that batch sampling with an inspection level of S-4 and an AQL, expressed in terms of defects per hundred units, shall be 1.0.

4.4.2.2 Intermediate inspection. The sample unit for this inspection shall be one actuator assembly, one complete tube assembly, link, bracket, or bridle, as applicable (see 3.2). Lots shall be expressed in units of components of one kind.

* 4.4.3 End item examination. Defects found during the examination shall be classified in accordance with the list shown in 4.4.3.1. The lot size shall be expressed in units of end items. The sample unit shall be one completely fabricated end item. The inspection level shall be I and the acceptable quality levels (AQLs) shall be 1.0 major and 6.5 total (major and minor combined) defects per hundred units.

* 4.4.3.1 End item visual examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of end items. The sample unit shall be one complete end item. The inspection level shall be 1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
General			
Finish, protective	Primary finish not as specified	X	
	Supplementary finish not as specified or tacky		X
	Any area of rust or corrosion	X	
	Flaky, pitted, blistered or otherwise loosened	X	
	Scratches, cuts or abrasion exposing base metal		X
	Foreign matter, the removal of which exposes base metal		X
Surface texture	Roughness exceeds limit on:		
	Surfaces 63 and finer	X	
	Surfaces 125 and rougher		X

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding	Any weld missing or not type and size specified	X	
	Any weld incomplete or cracked	X	
	Excessive spatter, undercut or not flush where specified		X
Stitching	Stitch type not as specified	X	
	Stitch pattern not as specified	X	
	One stitch per inch less than minimum specified		X
	Two or more stitches per inch less than minimum specified	X	
	Thread breaks or skipped stitches		X
	Ends of stitching not backstitched		X
	Loose tension, resulting in loosely exposed needle or bobbin thread; loops or knots; tight tension, resulting in tightness or puckering of material at stitch line		X
	Thread ends not trimmed		X
	Any row of stitching missing	X	
	Needle chews		X
	Hand tacking not as specified		X
	Thread breaks and skipped stitches overstitched less than 1 inch each end		X
	Stitching not carried over reinforcement as specified		X
General			
Construction and workmanship	Any component missing (excluding standard hardware not affecting function)	X	
	Any standard hardware component missing (not affecting function)		X
	Any standard hardware component wrong type or size	X	
	Any component cracked, fractured, malformed or misaligned	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
General			
Construction and workmanship (cont'd)	Any sharp edges or burrs (not otherwise covered)		X
	Any operation (not otherwise covered) omitted or not performed as specified	X	
Identification marking	Missing, not legible, incorrect or not permanent		X
Detailed			
Actuator	Any component of incorrect hand or not assembled as specified	X	
	Marking for "A" and "B" holes incorrect, missing or not legible	X	
	Nylon cord improperly routed or tied or ends not seared		X
Bridles	Any holes, cut or tear in webbing	X	
	Ends of webbing not seared or wax dipped as specified		X
	Any tear, cut or longitudinal split in keepers, or not wrapped as specified		X
Tubes	Do not telescope freely	X	
	"NO STEP" marking missing or incorrect		X

* 4.4.3.2 End item dimensional examination. The end item shall be examined for conformance to the applicable drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of end items. The sample unit shall be one complete end item. The inspection level shall be 1 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.4 Packing inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Marking (interior and exterior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, defective, or not as specified
Workmanship	Components not snug fit in container (loose)

4.4.5 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

Examine	Defect
Finished dimensions	Length, width or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location sequence or method of application.

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4.5 Methods of inspection.

4.5.1 Test, actuator proofload and function.

4.5.1.1 Equipment. The following equipment will be required for performance of this test:

(a) A tension device capable of applying a measured 9000 pound minimum load.

(b) A fixture upon which to mount the actuator. The fixture shall be constructed so as to permit free rotation of the actuator arm from the cocked position to approximately 10° beyond the unloaded position. A stop shall be provided at the latter position to prevent reverse loading of the arm torsion spring.

(c) A bridle, of construction similar to that shown on Drawing 11-1-1689, with which to couple the actuator to the tension device. Length to suit such connection.

(d) A locking rod to be used in place of the actuator safety pin. The rod should be of steel, 1/4 inch in diameter, approximately 2 feet long and having an "L" or "T" shaped hand grip at one end.

4.5.1.2 Performance. With the completely assembled actuator, either left or right hand, installed in the test fixture, the following operations shall be performed in sequence:

(a) Restrain the actuator arm from rotating and withdraw the safety pin. Place the safety pin on the opposite side of the actuator out of the way, and carefully rotate the actuator arm to the unloaded position at the fixture stop.

(b) Rotate the bridle retainer in the clear and slip the loop end of the bridle over the retainer. Return the retainer with attached bridle to its former position and cock the actuator. Lock the actuator in cocked position by installing the locking rod through the safety pin holes in both side plates.

(c) Secure the opposite end of the bridle to the tension device and apply a 9000 pound load to the actuator. Failure to conform to the requirements of 3.3.1 shall be cause for rejection.

(d) With the specified load still applied, unload the actuator by rapidly withdrawing the locking rod. Failure to conform to the requirements of 3.3.2 shall be cause for rejection.

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4.5.1.3 Fit and function. Each actuator shall be cycled manually through the operational cycle. There shall be no binding or interference with the designed function of the actuator arm, bridle retainer, bell crank arm and torsion and extension springs. In addition, each actuator shall be inspected for proper fit of the safety pin, C-5A pin and mounting bolts. Any nonconformance shall be cause for rejection of the actuator. Each tube assembly shall be inspected for proper fit of individual tubes within each other and absence of burrs or sharp edges at the leading edge of each tube. Any nonconformance shall be cause for rejection of the tube assembly or individual tube, as applicable.

4.5.1.4 Assembly. Each coupling assembly shall be examined for completeness and proper components. Any nonconformance shall be cause for rejection of the assembly.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Actuator assemblies, left hand and right hand. Each actuator shall be individually unit packed in a weather resistant grade fiberboard box conforming to PPP-B-636. Prior to unit packaging, the trigger arm of each actuator assembly shall be placed in the loaded (under tension) position and held in place with the safety pin (see 5.4.2 for special marking). Each box shall be closed in accordance with method IV as specified in the appendix of the box specification.

5.1.1.2 Link assembly. The link assembly shall be unit packed in a weather resistant grade fiberboard box conforming to PPP-B-636. The box shall be closed as specified in 5.1.1.1.

5.1.1.3 Bracket. Each bracket shall be individually unit packed in a weather resistant grade fiberboard box conforming to PPP-B-636. Prior to unit packaging, the bolts and washers of each bracket shall be detached and placed in a coin-type paper envelope or draw-string type cloth bag. The envelope or bag shall be securely closed and placed in the same box as the parent bracket. Each box shall be closed as specified in 5.1.1.1.

5.1.1.4 Tube assemblies. Each tube assembly grouping (consisting of a forward section, forward intermediate section, aft intermediate section, and aft section) shall be telescoped together to form its shortest length and then be unit packed in a weather resistant grade fiberboard box conforming to PPP-B-636. The box shall be closed as specified in 5.1.1.1.

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5.1.1.5 Bridle assembly. Each bridle assembly shall be compactly folded and then securely tied with twine conforming to type I or II, 8-ply of T-T-871. Each bridle assembly shall then be unit packed in a close-fitting 0.006-inch thick polyethylene bag conforming to type I or II, grade A, style 1 of PPP-B-26. Prior to or during the heat seal closure operation, excess air within the bag shall be expelled.

* 5.1.2 Commercial. The coupling assemblies shall be packed in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. One complete coupling assembly, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type, style A or I, grade A or B, type 2 load of PPP-B-601. The unit packs shall be neatly arranged within the container with the bridle assemblies placed on top of the boxed items. The unit packs shall be cushioned, blocked and braced, as applicable, to prevent movement while in transit. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-601, except the inspection shall be in accordance with 4.4.4.

* 5.2.2 Level B packing. One complete coupling assembly, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except that the shipping container shall conform to domestic type, style A or I, grade B, type 2 load of PPP-B-601, except the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Coupling assemblies, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), complete coupling assemblies, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in
MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready information.

5.4.2 Special marking. Each interior box containing an actuator assembly (see 5.1.1.1) shall have the following precautionary information legibly printed or stencilled upon the box:

"CAUTION:
TRIGGER ARM UNDER TENSION
USE CARE WHEN REMOVING
SAFETY PIN"

6. NOTES

6.1 Intended use. The coupling system is used in airdrop of cargo loads to effect transfer of the force developed by the extraction parachute system from the function of extracting the airdrop platform to that of deployment of the recovery parachute system. It differs from similar systems in that the extraction force is applied directly to the platform, rather than to the load.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Stock number.
(c) When a first article is required (see 3.1, 4.3 and 6.3).
(d) Selection of applicable levels of packaging and preservation (see 5.1 and 5.2).
(e) When palletization is required (see 5.3).

* 6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

* 6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specified instructions in all acquisition instruments, regarding arrangements for inspection and approval of the first article.

6.5 Conduct of tests. The supplier should observe reasonable precautions during conduct of tests to prevent injury to personnel performing testing.

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6.6 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies to these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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Preparing activity:

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Project No. 1670-A560

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-43899A(GL)	Coupling, Extraction Force Transfer, Platform

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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